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# Synchronized DNA cycling across a bacterial population

Leo Baumgart[1,2], William Mather[3,5] & Jeff Hasty[1,2,4,5]

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**A defining goal of synthetic biology is to engineer cells to coordinate tasks that often require precise temporal modulation of gene expression. Although a variety of relatively small gene circuits have been constructed and characterized, their logical combination into larger networks remains a central challenge. This is due primarily to the lack of compatible and orthogonal elements for predictable dynamic control of gene expression. As an alternative approach to promoter-level regulation, we explored the use of DNA copy number as a circuit control element. We engineered colony-wide DNA cycling in *Escherichia coli* in the form of plasmid copy number oscillations via a modular design that can be readily adapted for use with other gene circuitry. Copy number modulation is a generalizable principle that adds a layer of control to synthetic gene circuits, allowing dynamic regulation of circuit elements without requiring specially engineered promoters.**

The adaptation of DNA copy number in response to environmental pressures is a widespread mechanism found in natural systems. Copy number adjustments can be observed as relatively fixed changes, such as the evolution of ribosomal DNA tandem arrays, but also can drive rapid alterations in gene expression programs[1,2]. In synthetic biology, DNA copy number has typically been used to tune static gene expression levels. Recently, a synthetic circuit demonstrated plasmid amplification driven by changes in *E. coli* growth state as cultures approached saturation[3]. In this work we present strategies that allow both negative and positive plasmid copy number modulation in *E. coli* cells grown at a fixed density in continuous culture. We demonstrate that this can enable the rational design of synthetic circuits that harness plasmid copy number to dynamically control expression levels, in single genes or entire gene modules.

We initially investigated an approach for reversibly repressing gene expression by reducing copy number. We found that when a nuclease is expressed alongside a plasmid that contains the nuclease recognition sequence, the plasmid's copy number can be temporarily reduced below its natural levels. To characterize this effect, we constructed a ColE1 origin plasmid with a unique recognition sequence for the I-SceI endonuclease on the backbone and a constitutive promoter driving a sequence encoding a red fluorescent protein (RFP). On a second plasmid with the compatible p15A origin, we used an arabinose-inducible promoter to drive expression of the *SCEI* gene from *Saccharomyces cerevisiae*, which encodes the I-SceI protein (plasmid diagrams are shown in **Supplementary Fig. 1**). In *E. coli* cells transformed with both plasmids, the induction of I-SceI with arabinose resulted in a decrease of the ColE1 plasmid copy number, as evidenced by a reduction in RFP levels both in batch (**Fig. 1a**) and in continuous culture (**Fig. 1b**). We verified this result by qPCR (**Fig. 1a**) and found that the majority of linearized plasmids were quickly degraded (**Supplementary Fig. 2**). The expression of high levels of nuclease did not result in any obvious effect on cell size or growth, as assessed by single-cell microscopy (**Fig. 1b**, bottom). These experiments demonstrate that a targeted nuclease can be used to negatively regulate the expression of genes on a plasmid—even those driven by unregulated promoters.

We reasoned that controlled nuclease expression could enable the regulation of an entire module of genes and promoters when placed on a plasmid containing the cognate cut site. To demonstrate the utility of this novel mode of regulation, we constructed a synthetic gene oscillator that uses plasmid copy number repression by a nuclease as the sole negative feedback component. We adapted the architecture of the circuit (**Fig. 1c**) from a synthetic oscillator previously constructed by our group[4], using the *lux* quorum-sensing system from *Aliivibrio fischeri*[5]. In brief, LuxI catalyzes the production of a diffusible *N*-acyl homoserine lactone (AHL) molecule, which binds to the constitutively produced LuxR transcription factor and activates transcription from the *luxI* promoter, thereby forming a positive feedback loop. In the synthetic oscillator circuit, the *lux* quorum-sensing genes *luxI* and *luxR* are placed under their native bidirectional promoter on the ColE1 origin plasmid ('activator plasmid'), with an additional copy of this promoter driving the expression of a sequence encoding a green fluorescent protein (GFP), which can be used as a readout of the *luxI* promoter activation state. A second p15A origin plasmid ('repressor plasmid') contains a third copy of the *luxI* promoter driving expression of I-SceI, which targets and represses the activator plasmid. To facilitate fast protein-turnover dynamics, LuxI, GFP, and I-SceI all have an added ssrA tag targeting them for degradation by the native *E. coli* ClpXP protease[6]. When we grew *E. coli* with both of these plasmids in continuous culture in microfluidic cell chambers, we observed regular oscillations of GFP expression that were synchronized across the cells within each chamber (**Fig. 1d**,**e** and **Supplementary Video 1**; see **Supplementary Fig. 3** for the microfluidic chip design). This synthetic oscillator circuit demonstrates that nuclease-mediated copy

[1]Molecular Biology Section, Division of Biological Science, University of California, San Diego, La Jolla, California, USA. [2]BioCircuits Institute, University of California, San Diego, La Jolla, California, USA. [3]Departments of Physics and Biology, Virginia Polytechnic Institute and State University, Blacksburg, Virginia, USA. [4]Department of Bioengineering, University of California, San Diego, La Jolla, California, USA. [5]These authors contributed equally to this work. Correspondence should be addressed to J.H. (hasty@bioeng.ucsd.edu).

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number repression can override the strong positive feedback provided by the *lux* quorum-sensing module and is a powerful tool for controlling gene expression.

Because the synthetic oscillator circuit we constructed relies on induced copy number changes in one plasmid, we measured the copy number of the second plasmid in the system as well. It has been reported that for the ColE1 family plasmids, including p15A, copy number can be altered by changes in cell state, such as metabolic burden resulting from high rates of translation[7]. To investigate whether the copy number of p15A was affected over the course of a period of oscillation, we measured RFP expression from a constitutive promoter on the p15A repressor plasmid. We observed a small increase in RFP signal after each peak of GFP signal (**Fig. 2**), which suggested a slight temporary amplification in p15A copy number. In response to this observation, we considered whether repressor plasmid copy number modulation could have a dynamic role in oscillations.

To explore this further, we constructed a p15A plasmid for which *luxI* promoter activation directly amplified plasmid copy number. We took advantage of the native regulation of the p15A plasmid, which is composed of an antisense RNA system[8,9]. Overproduction of one of these RNA transcripts (RNAII) has been shown to lead to copy number amplification[10]. We removed the transcriptional terminator directly downstream of the *SCEI* gene on the repressor plasmid and replaced it with a second copy of the *luxI* promoter, such that transcription from these promoters drove into the p15A origin in the same direction as the RNAII promoter (**Fig. 2a**). To visualize the effect of transcription into the origin on plasmid copy number, we replaced the *SCEI* gene with an ssrA-tagged *gfp* gene that could serve as a reporter for *luxI* promoter activation. We used RFP expressed from a constitutive promoter on the same plasmid as a reporter for copy number. As expected, we observed that *luxI* promoter activation by exogenously added AHL, which manifested as an increase in GFP levels, led to amplification of p15A copy number and a transient increase in RFP (**Fig. 2b**). qPCR measurements showed a 2.3-fold ($P = 3 \times 10^{-8}$) amplification of p15A copy number after 90 min of induction with 450 nM AHL (**Supplementary Fig. 4**).

Next, we investigated the effects of this modified repressor plasmid on the oscillator dynamics. We imaged cells transformed with both the activator plasmid and the modified repressor plasmid, grown under the same conditions used in the previous microfluidic experiments. Using the RFP reporter, we found that the previously observed oscillations in repressor plasmid copy number were indeed amplified 1.5-fold ($P = 4 \times 10^{-7}$) in this modified oscillator circuit (**Fig. 2c**, bottom). The introduction of RNAII overexpression also led to a decrease in period and altered waveform compared with the first oscillator. To assess whether the use of a nuclease and RNAII overexpression adversely affects plasmid maintenance and thus the stability of this circuit, we imaged growing cells in a microfluidic device continuously for several days under constant antibiotic selection. We observed regular oscillations without apparent mutations or significant effects on cell growth (**Fig. 2d**,**e** and **Supplementary Video 2**).

We compared the two oscillator circuits to investigate whether the addition of engineered copy number amplification to the repressor plasmid could produce more robust oscillations. As a test for robustness with respect to microfluidic device geometries, we cultured cells of each strain in an alternative microfluidic device with cell chambers designed to have an approximately fivefold greater volume per chamber (**Supplementary Fig. 5**). When the chambers were seeded with the strain containing the second circuit modified to include engineered RNAII overexpression, we observed regular oscillations in these larger chambers as well (**Fig. 3a** and **Supplementary Video 3**). Furthermore,

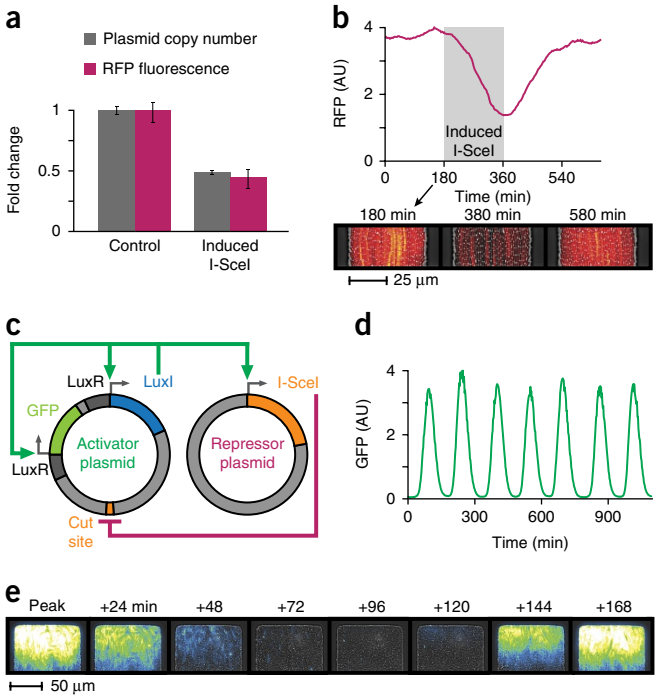



**Figure 1** A meganuclease (I-SceI) serves as a negative feedback element in this synthetic quorum oscillator by targeting the activator plasmid. (**a**) To demonstrate copy number modulation, we placed I-SceI under the control of an arabinose-inducible promoter, with I-SceI targeting a ColE1 origin plasmid that codes for constitutive RFP production. After 3 h of exponential growth in a flask containing arabinose, both RFP levels and ColE1 copy number decreased significantly ($P = 2 \times 10^{-4}$ for RFP, $P = 6 \times 10^{-12}$ for copy number). Plasmid copy number was quantified by qPCR with primers spanning the cut site. Data are shown as mean and s.e.m. We used an independent two-sample *t*-test for statistical analysis, with $n = 10$ replicate wells each for fluorescence measurements and qPCR. (**b**) Growth of the strain from **a** in a microfluidic cell chamber. RFP production dropped sharply after the induction of I-SceI by arabinose (gray shaded region), then recovered after the removal of arabinose. The images below the plot show a composite of phase contrast and RFP fluorescence at the indicated time points. AU, arbitrary fluorescence units (background subtracted). (**c**) The two-plasmid circuit: a ColE1 origin plasmid (activator plasmid) includes the *lux* quorum-sensing system from *A. fischeri*, which serves as a positive feedback loop for synchronized *luxI* promoter activation. A p15A origin plasmid (repressor plasmid) has the *luxI* promoter controlling expression of I-SceI, which targets the activator plasmid and reduces the copy number. LuxI, GFP, and I-SceI proteins are ssrA-tagged for ClpXP degradation. (**d**) Time series of average GFP signal from a representative chamber, showing regular synchronized oscillations produced by the circuit. The result is representative of seven replicate microfluidic chambers per experiment, with at least two replicate experiments for each strain. (**e**) A film strip showing a composite of phase contrast and GFP fluorescence produced by the oscillator circuit, covering one period of oscillation.

this modified circuit produced oscillations even when the growth medium was switched from rich lysogeny broth (LB) to minimal salts (M9) medium with glycerol (**Fig. 3b**). In contrast, we observed at best low-amplitude irregular oscillations in the larger microfluidic chambers when we used the original circuit with the unmodified repressor plasmid grown in either LB or M9 medium (**Supplementary Video 4** shows a side-by-side comparison). This suggests that modifying the circuit to incorporate both negative and positive DNA copy number regulation produces more robust oscillations across different culturing conditions.

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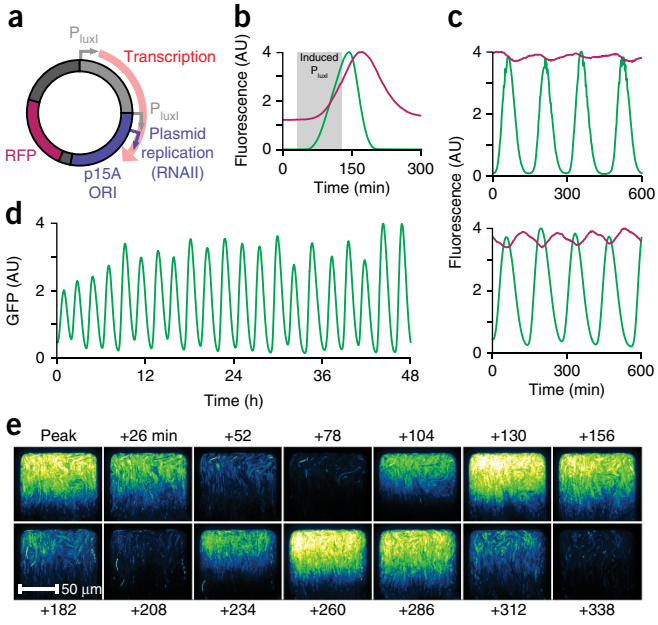



**Figure 2** Amplification of DNA copy number by engineered transcription into the origin of replication. (**a**) The arrangement of promoters allowing overproduction of RNAII. A second *luxI* promoter ($P_{luxI}$) replaces the terminator after the *SceI* gene, with both promoters facing the same direction as the native p15A promoter that primes plasmid replication (RNAII promoter). Constitutive RFP production reports on plasmid copy number. (**b**) To demonstrate the effect of *luxI* promoter activation on copy number, we replaced the *SceI* gene with *gfp*. Induction with 450 nM AHL for 90 min (shaded region) resulted in GFP production from the *luxI* promoter (green) and a concomitant increase in RFP signal (red), indicating plasmid copy number amplification. Note that GFP was ssrA-tagged for degradation, whereas RFP was untagged. (**c**) Small oscillations in RFP constitutively expressed from the p15A plasmid (red) were apparent even in the circuit without RNAII overproduction (top; circuit as shown in **Fig. 1c**). When the repressor plasmid was modified to include RNAII overproduction, these RFP oscillations were magnified 1.5-fold (bottom; $P = 4 \times 10^{-7}$). In both circuits, GFP (green) was expressed from the *lux* promoter on the activator plasmid. (**d**) Time series of average GFP signal in a representative cell chamber, showing regular synchronized oscillations produced by the circuit with RNAII overproduction. (**e**) A film strip of GFP fluorescence in a single chamber over approximately 2.5 periods of oscillation, produced by the same circuit as in **d**.

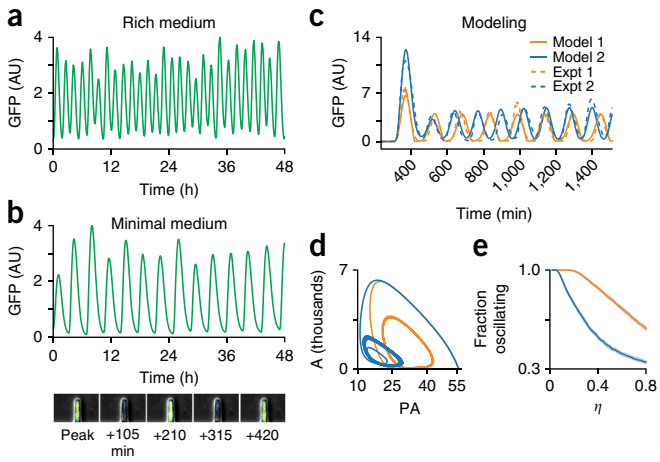



**Figure 3** Robustness and model analysis of the DNA feedback circuit. (**a**) Cells containing the circuit modified for repressor plasmid amplification produced regular oscillations in an alternative microfluidic device with taller chambers and a larger volume. (**b**) Observations from the same microfluidic device and bacterial strain from **a** cultured with minimal salts medium instead of rich medium. Oscillations remained regular but shifted to a longer period. The images at the bottom show a composite of phase contrast and GFP fluorescence in a single chamber over two periods of oscillation. (**c**) Trajectories that show good agreement between the mean GFP intensities from our two models and the corresponding mean GFP intensities averaged across single representative microfluidic traps (experiments (Expt) 1 and 2). Experiments have been aligned in time so that the first peaks have a similar time. (**d**) Projection of these oscillations onto LuxI concentration (A) and plasmid copy number for the activator (PA). The model predicts that plasmid copy number does not require excessive variation, which is important to avoid plasmid extinction. (**e**) A robustness analysis of our model fit as a function of parameter perturbation strength $\eta$, suggesting that our model is reasonably robust in a high dimensional parameter space (**Supplementary Note 1**). The curves represent the fraction of perturbed parameter sets that oscillated out of 100,000 parameter sets per $\eta$ value. The statistical error (sample s.d. of the batch mean) is indicated by gray shading, and was determined by bootstrapping over 100 independent batches of 1,000 parameter sets.

We integrated our observations into quantitative reaction network models for the circuit without RNAII overexpression feedback (model 1) and the circuit with RNAII overexpression feedback (model 2), which differed only in that model 2 included control of the repressor plasmid copy number. We based these models on degrade-and-fire models previously fit to experiments[4,11,12], including the effects of both delay in feedback[11] and proteolytic queuing[13] (details are presented in **Supplementary Note 1**). The models agreed well with experimental trajectories (**Fig. 3c**), and predicted that oscillation amplitudes for the plasmid copy number are small enough that plasmid extinction in cells is a rare event (**Fig. 3d**). A robustness analysis of the models showed that these oscillator designs are robust to general parameter variation (**Fig. 3e** and **Supplementary Fig. 6**). Curiously, we identified model 1 as more robust than model 2; this observation does not seem to be consistent with our experiments, but we believe that the more sinusoidal oscillations in **Figure 2d** as compared with the strong relaxation oscillations in **Figure 1d** placed model 2 in the vicinity of a Hopf bifurcation, which would be consistent with lower robustness for model 2. Additional investigation of quasi-1D scans of robustness supported the idea that model 2 is less robust and close to a bifurcation (**Supplementary Fig. 7**). We addressed the robustness experiments (**Fig. 3a**,**b**) by demonstrating that slightly perturbed parameters consistent with a change in trap geometry could lead to a situation in which only model 2 oscillates, as in the experiment (**Supplementary Fig. 8**). This observation indicated that the robustness of the circuits may be dependent on trap geometry.

The engineered interactions discussed previously are sufficient to explain oscillations, but we also considered the effect of implicit interactions arising from the limited abundance of transcription factors. When transcription factors are not present in a large excess, the ratio of binding sites to binding proteins becomes relevant to gene expression. An amplified DNA copy number necessarily implies an increased number of binding sites, an effect that is multiplied when a single DNA copy contains multiple binding sites. In the oscillator circuits described in this work, positive feedback is sustained only when LuxR sufficiently binds to and activates the *luxI* promoter located upstream of the *luxI* gene. Thus we reasoned that this positive feedback could be effectively interrupted if sufficient decoy binding sites were supplied to titrate LuxR away from this promoter. We demonstrated this by modifying a circuit to remove the effect of the nuclease and to uncouple quorum sensing from repressor plasmid

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amplification, and we observed that LuxR transcription factor titration counteracted ongoing positive feedback from the activator plasmid (**Supplementary Fig. 9**). This indirect interaction due to transcription factor titration points toward future opportunities to control DNA copy number modulation in applications.

The original genetic clock[14] and toggle switch[15] circuits firmly established the engineering pillar of the field of synthetic biology. While the two designs used similar forms of transcriptional regulation, they were also prescient with regard to the modern practice of parsing gene circuits into 'analog' and 'digital' components[16–18]. Subsequently, analog clocks evolved into platforms for exploring the synchronization of gene circuits in[4,19] and between bacterial colonies[20,21], whereas digital logic was generalized to complex intracellular algorithms[22–29] and memory storage[30,31]. Our work establishes a framework for the engineering of a DNA 'master clock' at the colony level that can serve to coordinate digital subprocessing within single cells. More broadly, our results demonstrate how DNA copy number modulation can be used as a general tool for controlling gene expression in synthetic biology.

## METHODS

Methods, including statements of data availability and any associated accession codes and references, are available in the online version of the paper.

*Note: Any Supplementary Information and Source Data files are available in the online version of the paper.*


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### AUTHOR CONTRIBUTIONS

L.B. and J.H. contributed to the circuit design and development of the project. L.B. constructed the plasmids and strains, conducted the experiments, and analyzed results. W.M. and J.H. conducted the mathematical analysis and computational modeling. L.B. and W.M. prepared the figures, and L.B., W.M., and J.H. prepared the manuscript.

### COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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## ONLINE METHODS

**Strains and culturing.** All plasmids were constructed by Gibson assembly followed by transformation into Mach1 (Invitrogen) chemically competent *E. coli*. Plasmids were verified by Sanger sequencing before transformation into *E. coli* strain MG1655. For strains containing quorum-sensing constructs, growth on plates was limited to a maximum of 10 h, and 0.2% (wt/vol) glucose was added to all plates and media used during cloning and transformation to prevent auto-induction. All experiments were carried out in the MG1655 strain with the appropriate antibiotics: 50 μg/ml kanamycin for strains containing ColE1 origin plasmids, and 34 μg/ml chloramphenicol for strains containing p15A origin plasmids. For the experiments with strain LABS1 involving induction of I-SceI with arabinose, cells were cultured in low-density conditions that prevented auto-induction. The AHL inducer used in experiments was 3-oxo-C6-HSL. A detailed description of all strains is available in **Supplementary Table 1**, and the corresponding plasmid diagrams are shown in **Supplementary Figures 1** and **9**. Plasmid sequence accession numbers are provided in the "Data availability" section. The doubling time of oscillator strains LABS2 and LABS4 was measured in a 96-well plate in rich LB medium with antibiotics as 20.7 min ($n = 6$ wells for each strain; s.e.m. = 0.2 min).

**Microfluidics and microscopy.** For each experiment, the appropriate *E. coli* strain was seeded from a −80 °C glycerol stock into 3 ml of LB supplemented with 0.2% (wt/vol) glucose, the appropriate antibiotics, and 0.075% (wt/vol) Tween 20. After growth for 8–12 h at 37 °C in a shaking incubator, the culture was diluted 100-fold into 3 ml of the same medium and grown for one additional hour. This culture was concentrated by centrifugation at 5,000*g* for 1 min and resuspended in 3 μl of the same medium. Cells were loaded by degas-driven flow into microfluidic chambers via the waste port. Microfluidic experiments were conducted with media supplemented with appropriate antibiotics and 0.075% (wt/vol) Tween 20. Experiments involving *lux*-mediated positive feedback were done with media containing a low background concentration of AHL, which was also required to induce oscillations: 5 nM AHL for experiments using rich LB medium, and 1 nM AHL for experiments using minimal medium. The minimal medium used was M9 salts medium supplemented with 0.4% (vol/vol) glycerol, 0.2% (wt/vol) casamino acids, and 1 μM thiamine. For experiments involving induction with chemical inducers, the medium source was switched manually at the device inlet at the times indicated in figures. Time-lapse images were acquired on a Nikon TI microscope fitted with a Lumencor SOLA SE light engine for fluorescence imaging. Average fluorescence values across each chamber were determined with Fiji software[32]. A single baseline fluorescence value was measured from a region outside the chamber for each time series and used for background subtraction. Schematics of the microfluidic devices are shown in **Supplementary Figures 3** and **5**.

For comparison of the oscillation amplitude of RFP in the two circuits, we de-trended RFP time series by subtracting a moving average over a window equal to the period of each circuit (152 min for the first circuit, 128 min for the modified circuit). De-trended time series were used to measure peak-to-trough amplitudes for each time series, which were averaged and normalized by the mean RFP fluorescence for the time series. Statistical significance was determined by independent two-sample *t*-test, using mean-adjusted average amplitude values for each time series with $n = 40$ peaks from seven time series for the first circuit and $n = 42$ peaks from six time series for the modified circuit, with d.f. = 80.

**Relative copy number determination by qPCR and fluorescence.** A culture was seeded from a −80 °C glycerol stock into 3 ml of LB medium supplemented with 0.2% (wt/vol) glucose. After 12 h of growth at 37 °C in a shaking incubator, the culture was diluted 100-fold into fresh LB medium without glucose and grown for an additional 2 h to ensure that cells were in the exponential growth phase. The culture was then diluted into 250-ml flasks, each containing 50 ml of prewarmed LB, and chemical inducer was added: 0.2% (wt/vol) arabinose for copy number repression with strain LABS1 (sterile water for control), and 450 nM AHL for copy amplification with strain LABS3 (DMSO for control). After growth in the presence of inducer (3 h with arabinose for LABS1, 1.5 h with AHL for LABS3), 100-μl samples were taken from each flask for qPCR, heated at 95 °C for 10 min, and then immediately frozen at −20 °C for subsequent qPCR analysis, as described by Škulj *et al.*[33].

For fluorescence measurements, cultures were normalized to the same $OD_{600}$ and pipetted into 10 replicate wells in a 96-well plate, and readings were taken with a Tecan Infinite M200Pro plate reader. A culture of wild-type *E. coli* strain MG1655 at the same $OD_{600}$ was used to measure background fluorescence signal, using the mean of ten replicate wells. Cultures were in the exponential growth phase at the time of sampling, with $OD_{600}$ readings of 0.1–0.2. Statistical significance for RFP fluorescence measurements was determined by independent two-sample *t*-test, with d.f. = 18.

qPCR primers were validated in a five-point template dilution series to ensure that the amplification efficiency for each pair was >90%. The genomic reference primers chosen were as described by Škulj *et al.*[33]. Three sets of primers were designed, targeting the ColE1 plasmid in different locations: primer set A with one primer on either side of the I-SceI cut site, primer set B beginning 147 bp from the cut site in the kanamycin-resistance gene, and primer set C beginning 2,722 bp from the cut site in the *gfp* gene. The same genomic reference primer set and primer set C were also used in a separate experiment to measure p15A plasmid amplification. qPCR primer sequences are presented in **Supplementary Table 2**. For relative copy number determination, cell culture samples were thawed, diluted 2,000-fold in sterile water, and added at a ratio of 2:3 to a master mix consisting of appropriate primers (500 nM final concentration) added to Bio-Rad iQ SYBR Green Supermix. Each reaction condition was pipetted into 10 replicate wells in a 96-well PCR plate, and threshold values were determined with a Bio-Rad CFX Connect Real-Time PCR detection system with the associated software. Melt curves were examined for each reaction well to verify the presence of a single amplification product. Relative copy numbers were calculated via the $\Delta\Delta C_t$ method, and error bars were calculated by propagation of the s.e.m. $C_t$ values. Statistical significance for $\Delta C_t$ values was determined by independent two-sample *t*-test with the mean $\Delta C_t$ values and associated propagated error with d.f. = 18.

**Quantitative modeling.** Model creation and fitting were done with the package COPASI[34] and custom Python scripts. Details concerning the quantitative modeling appear in **Supplementary Note 1**.

**Data availability.** Complete annotated sequences for both repressor plasmids and the activator plasmid are available in GenBank under accession numbers MF033067, MF033068, and MF033069. Plasmids, bacterial strains, and modeling files and scripts are available from the corresponding author upon request.


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